# interpolate_mev_to_grid.R

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`interpolate_mev_to_grid` *Interpolate 2-D Spatial MEVs to Prediction Grid*

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### Description

Uses Inverse Distance Weighting (IDW, power = 2) to approximate Moran Eigenvector Map (MEM) values from training-site locations to arbitrary prediction locations, then scales the result using training spatial scale attributes so that the interpolated predictors are on the same scale as those seen during model fitting.

### Usage

```
interpolate_mev_to_grid(
  data_coords_projected_train = NULL,
  data_mev_core = NULL,
  data_coords_projected_pred = NULL,
  spatial_scale_attributes = NULL,
  chunk_size = 5000L
)
```

### Arguments

`data_coords_projected_train`
: A data frame with ‘coord_x_km‘ and ‘coord_y_km‘ columns and ‘dataset_name‘ as row names, as returned by ‘project_coords_to_metric()‘. One row per unique training site.

`data_mev_core`
: A data frame with unscaled MEV columns (‘mev_1‘, ‘mev_2‘, ...) and ‘dataset_name‘ as row names, as returned by ‘compute_spatial_mev()‘.

`data_coords_projected_pred`
: A data frame with ‘coord_x_km‘ and ‘coord_y_km‘ columns and arbitrary row names identifying prediction locations (e.g. ‘"grid_1"‘, ‘"grid_2"‘), as returned by ‘project_coords_to_metric()‘.

spatial_scale_attributes
: Optional named list of ‘"scaled:center"‘ and ‘"scaled:scale"‘ attributes per MEV column, as returned by ‘scale_spatial_for_fit()‘ in the ‘spatial_scale_attributes‘ element. Used to bring interpolated MEV values onto the same scale as the training spatial predictors. When ‘NULL‘, unscaled interpolated values are returned for fold-local scaling.

chunk_size
: Positive integer limiting the number of prediction locations included in one distance matrix. Default is ‘5000L‘.

## Details

MEMs are eigenvectors of the spatial connectivity matrix at training sites and cannot be analytically evaluated at new locations. IDW (power = 2) with a small epsilon (1e-10) to prevent division-by-zero provides a smooth spatial interpolation.

This function handles the **2-D spatial case** only (x_km, y_km). For models fitted with ‘spatial_mode = "spatiotemporal"‘ use ‘interpolate_st_mev_to_grid()‘ instead.

## Value

A data frame with the same row names as ‘data_coords_projected_pred‘ and one column per MEV (names matching ‘data_mev_core‘). Columns are scaled when ‘spatial_scale_attributes‘ is supplied and otherwise remain unscaled.

## See Also

[compute_spatial_mev()], [interpolate_st_mev_to_grid()], [project_coords_to_metric()], [scale_spatial_for_fit()]

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